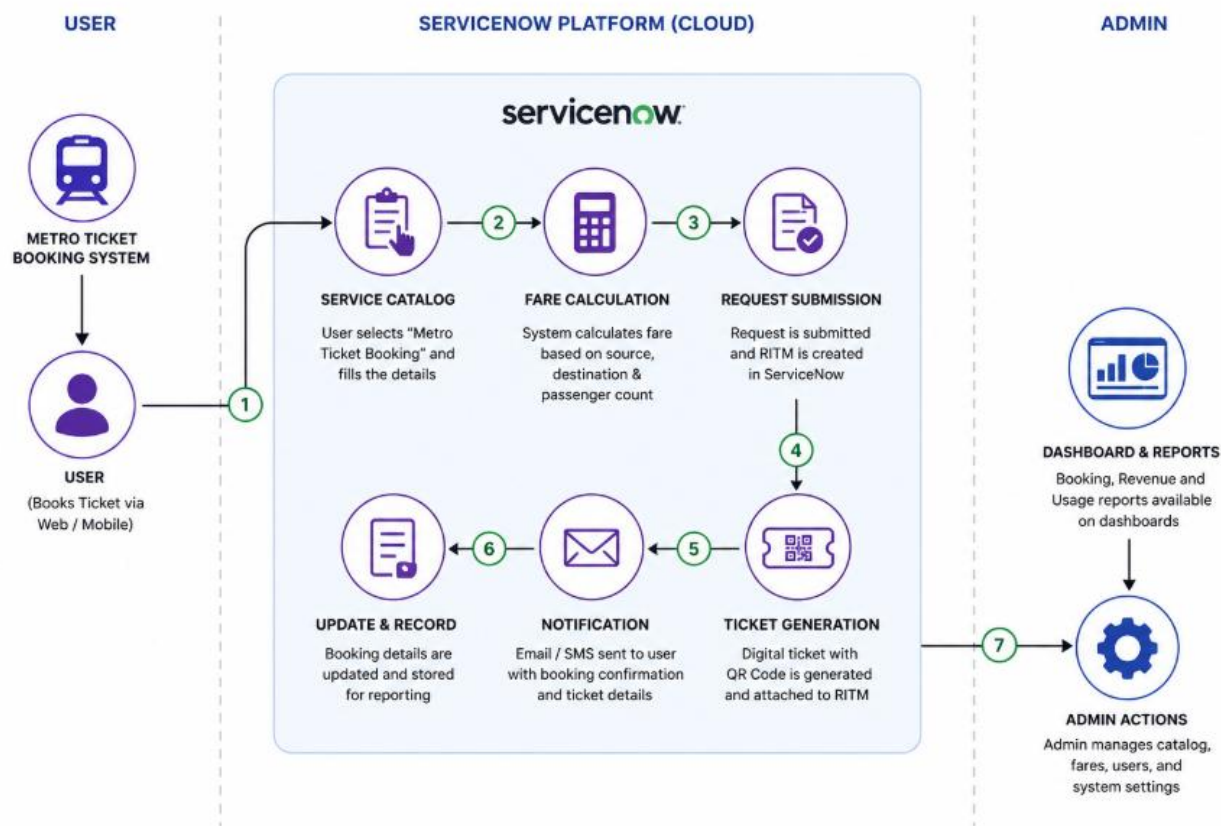


## Project Design Phase-II Technology Stack (Architecture & Stack)

|               |                                              |
|---------------|----------------------------------------------|
| Date          | 30-06-2026                                   |
| Team ID       | 24A55A4403                                   |
| Project Name  | Metro Ticket Generating System in ServiceNow |
| Maximum Marks | 4 Marks                                      |

### Technical Architecture:



**Table-1 : Components & Technologies:**

| S.No | Component                       | Description                                               | Technology                                                     |
|------|---------------------------------|-----------------------------------------------------------|----------------------------------------------------------------|
| 1.   | User Interface                  | Interface for passengers to book metro tickets            | ServiceNow Service Catalog, Portal UI                          |
| 2.   | Application Logic-1             | Fare calculation based on source and destination stations | Catalog Client Scripts (JavaScript)                            |
| 3.   | Application Logic-2             | Server-side fare retrieval and validation                 | Script Includes, GlideAjax                                     |
| 4.   | Application Logic-3             | Automated ticket booking workflow and notifications       | Flow Designer, Business Rules                                  |
| 5.   | Database                        | Stores metro stations, fares, and booking information     | ServiceNow Tables (u_metro_station, u_metro_fare, sc_req_item) |
| 6.   | Cloud Database                  | Cloud-based data storage managed by ServiceNow            | ServiceNow Cloud Database                                      |
| 7.   | File Storage                    | Stores QR codes, ticket attachments, and records          | ServiceNow Attachments                                         |
| 8.   | External API-1                  | QR Code generation for digital tickets                    | QR Code Generator API / Scripted QR Generation                 |
| 9.   | External API-2                  | Email notification service for ticket confirmation        | ServiceNow Notification Engine                                 |
| 10.  | Machine Learning Model          | Ticket booking trends, revenue, and usage reports         | ServiceNow Reports & Dashboards                                |
| 11.  | Infrastructure (Server / Cloud) | Application hosting and execution environment             | ServiceNow SaaS Cloud Platform                                 |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                                                                                                                    | Technology                                                   |
|------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1.   | Open-Source Frameworks   | Client-side scripting and customization used for fare calculation and form validation                                                          | JavaScript, GlideAjax                                        |
| 2.   | Security Implementations | Role-based access control for administrators and users, authenticated access through ServiceNow platform, secure data storage and transmission | ServiceNow ACLs, User Roles, HTTPS, IAM Controls             |
| 3.   | Scalable Architecture    | ServiceNow's cloud-based multi-tenant architecture supports growing users and ticket bookings without infrastructure changes                   | ServiceNow SaaS Architecture, Multi-Tenant Cloud Platform    |
| 4.   | Availability             | High availability through ServiceNow cloud infrastructure with continuous monitoring and managed services                                      | ServiceNow Cloud Infrastructure, Redundant Data Centers      |
| 5.   | Performance              | Fast ticket booking process through server-side processing, optimized database queries, and cloud-hosted services                              | GlideAjax, Script Includes, ServiceNow Database Optimization |

**References:**

<https://www.servicenow.com/docs/r/servicenow-platform/service-catalog/service-catalog.html>

<https://www.servicenow.com/docs/r/application-development/building-applications/flow-designer.html>

[https://www.servicenow.com/docs/r/api-reference/c\\_GlideAjaxAPI.html](https://www.servicenow.com/docs/r/api-reference/c_GlideAjaxAPI.html)

<https://developer.servicenow.com>

<https://www.servicenow.com/community/>